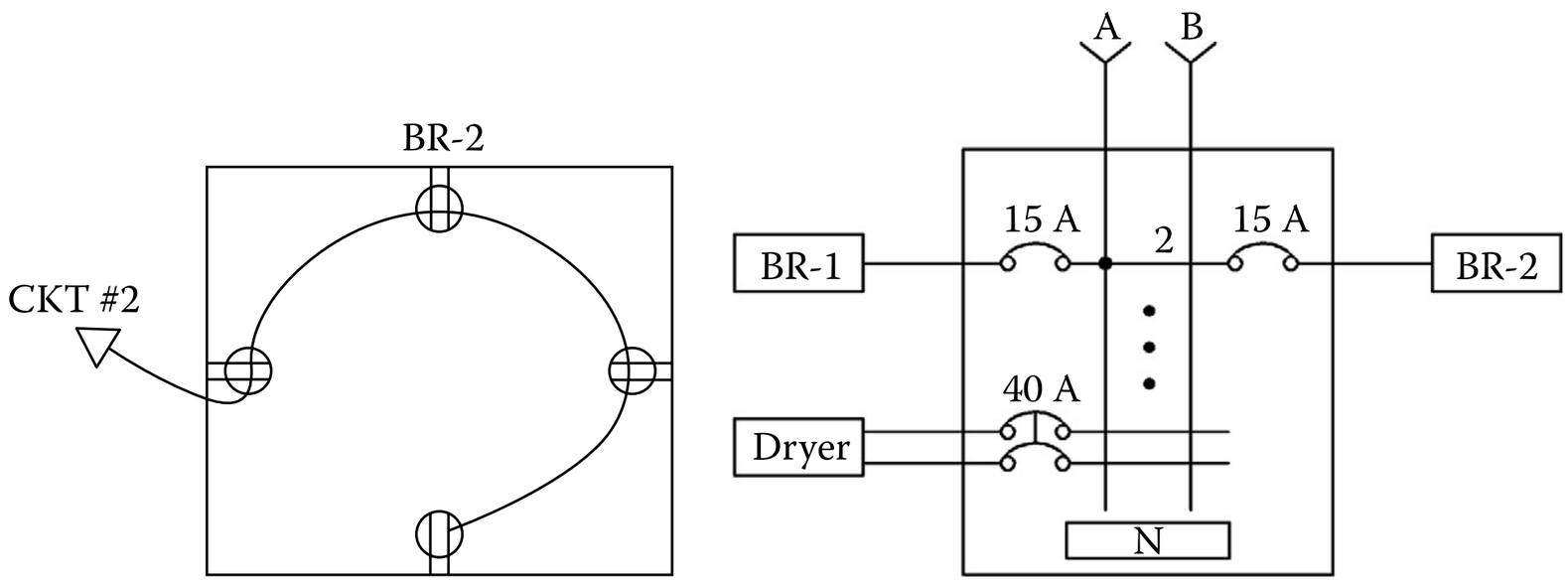


FIGURE 7.4
Identification of branch circuit serving a bedroom (BR-2) both (a) in a circuiting layout and (b) in a panelboard schedule.

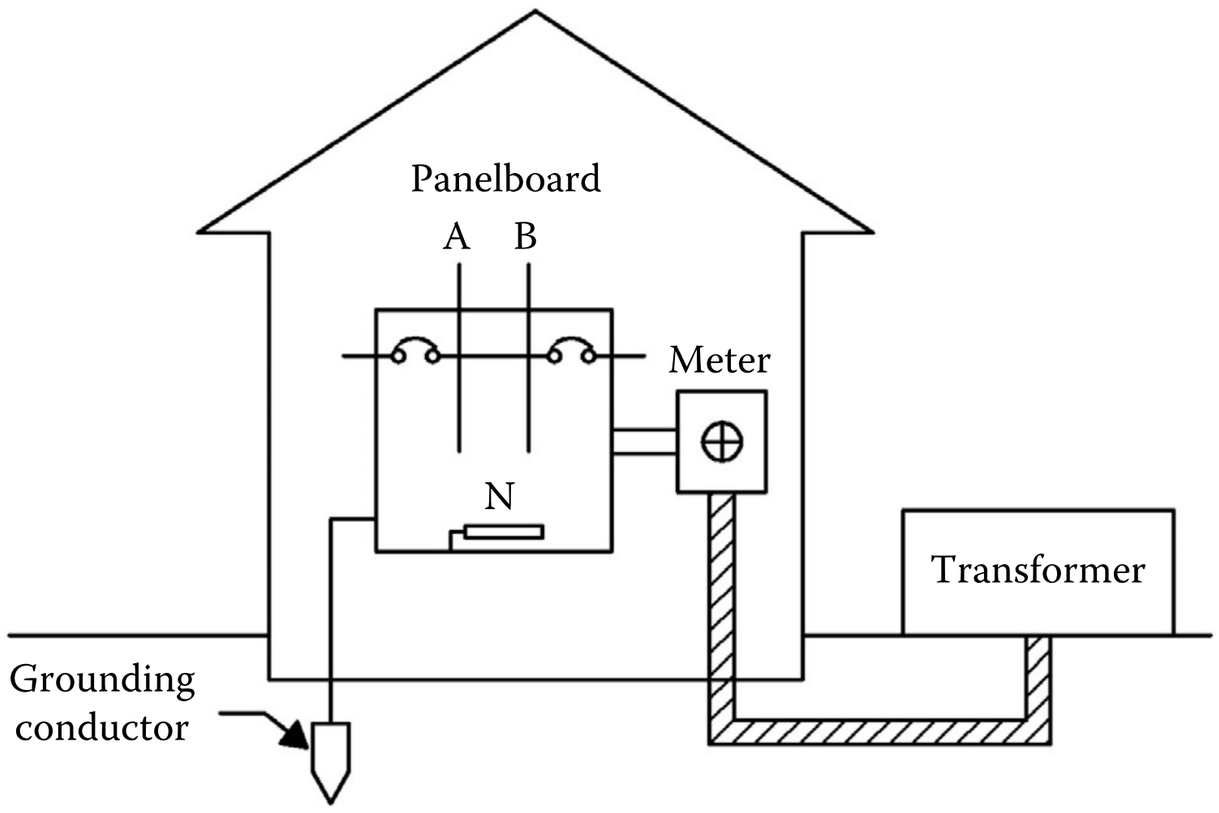


FIGURE 7.5
Typical grounding configuration for a dwelling.

7.3 Main Service Entrance Design

Service entrance feeder is a conductor or a set of conductors that serves the entire dwelling or the residential building. The size of the service entrance feeder can be determined using either the standard method or the optional method outlined in the NEC. The circuit breaker protecting the panelboard is called the main circuit breaker (main CB) and is often located within the panel and defines the rating of the panelboard. Designing the service entrance includes determining the size of both the conductor and the main CB, as illustrated in Figure 7.6.

7.3.1 Standard Method for Sizing Service Entrance

In order to determine the service entrance size, all electrical loads for the dwelling should be obtained. These loads include lighting, receptacles, fixed appliances, special appliances, and heating/cooling systems. According to NEC, two calculation methods can be utilized

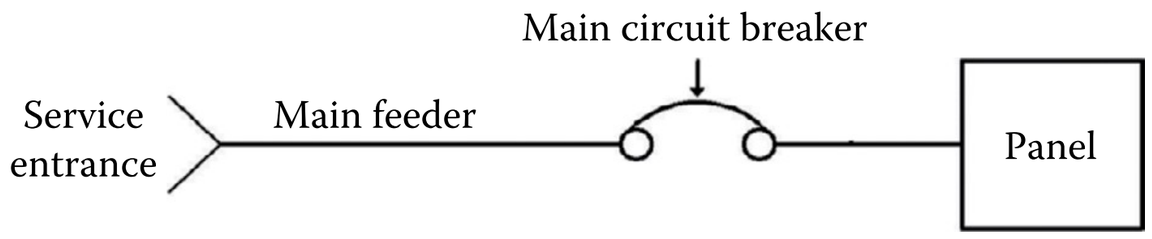


FIGURE 7.6
Sizing the main feeder conductor and main circuit breaker are part of the design of service entrance for dwellings.

to determine the phases (A and B for one-phase systems and A, B, and C for three-phase systems) and the neutral (N):

- Standard method: can be used to size any phase and neural wires (A, B, N for one-phase systems and A, B, C, N for three-phase systems).
- Optional method: can be used to size only the phase wires (A, B for one-phase systems and A, B, C for three-phase systems) in the case of the main CB ≥ 100 A.

In this section, the standard method calculation details for estimating the demand loads for the service entrance feeder are described.

7.3.1.1 Estimation of the Demand Load for the Phases (A, B, and, if Needed, C)

1. General loads (lighting/general receptacles)
   a. General load = 3 VA/ft² (or 33 VA/m²) * total livable building floor area
   b. Kitchen load = 2 branch circuits * 1500 VA/circuit
      i. Each branch circuit has to be rated at 20 A.
   c. Laundry load = 1 branch circuit * 1500 VA/circuit
      i. The branch circuit has to be rated at 20 A.

   Then, the demand load associated with general lighting and general-use receptacles, DL_{GL}, can be estimated as follows:

$$DL_{GL} = 100\% \text{ of first } 3,000\,\text{VA} + 35\% \text{ of remainder up to } 120,000\,\text{VA} + 25\% \text{ of the remainder over } 120,000\,\text{VA} \quad (7.1)$$

2. Fixed appliances load: apply demand factors (*DF*s) to the actual connected load (*CL*) for all the fixed appliances that served in the building using the following equation:

$$DL_{FA} = DF \times CL_{FA} \quad (7.2)$$

   with
   a. *DF* = 100% if the total number of fixed appliances ≤3
   b. *DF* = 75% if the total number of fixed appliances ≥4

3. Special appliances load: Typically, dryers, ranges, space heating, and cooling systems are considered special appliances and their demand loads can be estimated according to NEC guidelines.
   a. Table 7.1 should be used to estimate the demand loads for electrical dryers.
   b. Table 7.2 should be used to estimate the demand loads for electrical ranges and ovens.
   c. A demand factor of 100% should be applied to only the largest of the space heating or cooling system serving the dwelling to determine the demand loads associated with air conditioning systems for residential buildings.
   d. If motor is present, a demand factor of 25% should be applied to the largest motor load associated typically to one of the fixed appliances.
4. Total demand load: the total demand load, DL_{tot}, in VA can be simply estimated by summing the demand loads estimated in the previous three steps according to the following equation:

$$DL_{tot} = DL_{GL} + DL_{FA} + DL_{SA} \tag{7.3}$$

Then, the total load can be expressed in terms of amperage using the following expression:

$$I_{tot} = \frac{DL_{tot}}{E_L} \text{ for one-phase systems } (\text{such as } 120/240\text{V}) \tag{7.4}$$

$$I_{tot} = \frac{DL_{tot}}{\sqrt{3}E_L} \text{ for three-phase sytems } (\text{such as } 208\text{Y}/120\text{V})$$

7.3.1.2 Estimation of the Demand Load for the Neutral (N)

The same procedure outlined for estimating the demand load for the phases should be utilized, but only the one-phase loads should be considered, except the loads associated with ranges and dryers. Indeed, these two loads estimated using Tables 7.1 (dryers) and 7.2 (ranges) should be considered with an additional derating factor of 70%.

Example 7.1 illustrates the application of Tables 7.1 and 7.2 to determine the loads associated with dryers and ranges, while Example 7.2 utilizes for a specific house the standard method to estimate the demand load for both the phases and neutral.

Example 7.1

Problem

Determine the demand loads associated to electrical dryers and ranges for the following cases:

Case 1: six dryers, each rated at 5 kVA
Case 2: one range rated at 12 kVA
Case 3: one range rated at 9 kVA
Case 4: one range rated at 2.5 kVA
Case 5: one range rated at 27 kVA
Case 6: three ranges rated at 10, 14, and 19 kVA

TABLE 7.1

Demand Factors for Electrical Dryers

Number of Dryers	Demand Factor (%)
Less than 4	100.0
5	85.0
6	75.0
7	65.0
8	60.0
9	55.0
10	50.0
11	47.0
12	46.0
13	45.0
14	44.0
15	43.0
16	42.0
17	41.0
18	40.0
19	39.0
20	38.0
21	37.0
22	36.0
23	35.0
24	34.5
25	34.0
26	33.5
27	33.0
28	32.5
29	32.0
30	31.5
31	31.0
32	30.5
33	30.0
34	29.5
35	29.0
36	28.5
37	28.0
38	27.5
39	27.0
40	26.5
41	26.0
42	25.5
More than 43	25.0

Note: The demand load for one dryer is the largest of its nameplate rating and 5000 VA. This table is adapted from NEC-2014 Table 220.54 (NEC, 2014).

TABLE 7.2

Demand Loads and Demand Factors for Electrical Ranges and Ovens

Number of Appliances	Demand Factors (%)		Column C Maximum Demand (kW) (for 8.75–12 kW Ratings)
	Column A (Less than 3.4 kW Rating)	Column B (for 3.5–8.75 kW Ratings)	
1	80	80	8
2	75	65	11
3	70	55	14
4	66	50	17
5	62	45	20
6	59	43	21
7	56	40	22
8	53	36	23
9	51	35	24
10	49	34	25
11	47	32	26
12	45	32	27
13	43	32	28
14	41	32	29
15	40	32	30
16	39	28	31
17	38	28	32
18	37	28	33
19	36	28	34
20	35	28	35
21	34	26	36
22	33	26	37
23	32	26	38
24	31	26	39
25	30	26	40
26–30	30	24	15 kW + 1 kW for each range
31–40	30	22	15 kW + 1 kW for each range
41–50	30	20	25 kW + 1 kW for each range
51–60	30	18	25 kW + 1 kW for each range
61 and over	30	16	25 kW + 1 kW for each range

Notes: This table is adapted from NEC-2014 Table 220.55 (NEC, 2014).

1. For ranges individually rated more than 12 kW but no more than 27 kW, the maximum demand in Column C shall be increased by 5% for each additional kW of rating or major fraction thereof by which the rating of individual ranges exceeds 12 kW.
2. For ranges individually rated more than $8^{3/4}$ kW and of different ratings, but none exceeding 27 kW, an average value of rating shall be calculated by adding together the ratings of all ranges to obtain the total connected load (using 12 kW for any range rated less than 12 kW) and dividing by the total number of ranges. Then, the maximum demand in Column C shall be increased by 5% for each additional kW of rating or major fraction thereof by which this average value exceeds 12 kW.
3. For ranges individually rated more than $1^{3/4}$ kW but no more than $8^{3/4}$ kW, it is permissible instead of using the method provided in Column C to add the nameplate ratings of all household cooking appliances rated more than $1^{3/4}$ kW but no more than $8^{3/4}$ kW and multiply the sum by the demand factors specified in Column A or Column B for the given number of appliances. Where the rating of the cooking appliances falls between Column A and Column B, the demand factors for each column should be applied to the appliances of that column, and the results added together.

Solution

Using Tables 7.1 for dryers and 7.2 for ranges, the demand loads for all the cases can be estimated as summarized here:

Case 1: DL = 0.75 * 6 * 5 kVA = 22.5 kVA (based on Table 7.1 demand factor for six dryers)
Case 2: DL = 8 kVA (based on Table 7.2, Column C)
Case 3: DL = 8 kVA (based on Table 7.2, Column C)
Case 4: DL = 0.80 * 2.5 kVA = 2 kVA (based on Table 7.2, Column A)
Case 5: Using Note 1 and Column C of Table 7.2, first the demand factor, *DF*, is estimated:

$$DF = 1 + (27 - 12) * 0.05 = 1.75$$

Then, the demand load can be determined:

$$DL = 1.75 * 8 = 14 \text{ kVA}$$

Case 6: Using Note 2 of Table 7.2, the following steps are outlined to estimate the demand load of three ranges with different ratings:
Step 1: Increase 10 to 12 kVA.
Step 2: Take average of all three ratings = (12 + 14 + 19)/3 = 15 kVA.
Step 3: Estimate the demand factor: *DF* = 1 + (15–12) * 0.05 = 1.15.
Step 4: Calculate the demand load: *DL* = 1.15 * 14 = 16.1 kVA.

Example 7.2

Problem
Determine the demand load for both phases A and B as well as the neutral for a service entrance feeder serving 1600 ft^2 (150 m^2) house that has the following electrical loads:

400 VA disposable (120 V), 800 VA compactor (120 V), 4.5 A blower motor (120 V), 1,400 VA dishwasher (120 V), 4,800 VA water heater (240 V), 5,000 VA dryer (240 V), 12,000 VA range (240 V), and 20,000 VA space heaters (240). There are three separately controlled heaters in the dwelling.

Use the NEC standard method to determine the demand loads for A, B, and N.

Solution

First the load for the phases A and B are estimated considering all the loads. Then, the neutral is sized using only phase-to-neutral loads (i.e., 240) except as noted by NEC the loads associated with the range and the dryer.

Phases A and B demand load

1. General lighting and receptacles load:
   General load = 3 VA/ft^2 * 1600 ft^2 = 4800 VA
   Kitchen load = 2 circuits * 1500 VA/circuit = 3000 VA
   Laundry load = 1 circuit * 1500 VA/circuit = 1500 VA
   Thus, the total connected load = 9300 VA
   The demand load for general lighting and receptacles load:

$$DL_{GL} = 100\% * 3000 \text{ VA} + 35\% * (9300 - 3000) \text{ VA} = 5205 \text{ VA}$$

2. Fixed appliances load:
   Disposal (use rated VA) = 400 VA
   Compactor (use rated VA) = 800 VA

Blower motor = 4.5 A * 120 V = 540 VA
Dishwasher (use rated VA) = 1400 VA
Water heater (use rated VA) = 4800 VA
Thus, the total connected load = 7940 VA
The demand load for the fixed appliances load:

$$DL_{FA} = 75\% * 7940 \text{ VA} = 5955 \text{ VA}$$

3. Special appliances load:
   Range (use Table 7.2) = 8000 VA
   Dryer (use Table 7.1) = 5000 VA
   Space heaters (use rated VA) = 20000 VA
   Largest motor (25% * compactor VA) = 200 VA
   Thus, the demand load for the special appliances load:

$$DL_{SA} = 33,200 \text{ VA}$$

The total demand load for phases A and B:

$$DL_{tot,A-B} = 5{,}205 \text{ VA} + 5{,}955 \text{ VA} + 33,200 \text{ VA} = 44{,}360 \text{ VA}$$

Therefore, the demand load in terms of current for phases A and B:

$$I_{tot,A-B} = 44,360 \text{ VA}/240 \text{ V} = 184.8 \text{ A}$$

Neutral N demand load

1. General lighting and receptacles load:
   General load = 3 VA/ft² * 1600 ft² = 4800 VA
   Kitchen load = 2 circuits * 1500 VA/circuit = 3000 VA
   Laundry load = 1 circuit * 1500 VA/circuit = 1500 VA
   Thus, the total connected load = 9300 VA
   The demand load for general lighting and receptacles load:

$$DL_{GL} = 100\% * 3000 \text{ VA} + 35\% * \left(9300 - 3000\right)\text{VA} = 5205 \text{ VA}$$

2. Fixed appliances load: only 120 V appliances are considered:
   Disposal (use rated VA) = 400 VA
   Compactor (use rated VA) = 800 VA
   Blower motor = 4.5 A * 120 V = 540 VA
   Dishwasher (use rated VA) = 1400 VA
   Thus, the total connected load = 3140 VA
   The demand load for the fixed appliances load:

$$DL_{FA} = 75\% * 3140 \text{ VA} = 2355 \text{ VA}$$

3. Special appliances load: only 120 V appliances except range and dryer loads
   Range (use Table 7.2 and a derating factor of 70%) = 70% * 8000 VA = 5600 VA
   Dryer (use Table 7.1 and a derating factor of 70%) = 70% * 5000 VA = 3500 VA
   Largest motor (25% * compactor VA) = 200 VA
   Thus, the demand load for the special appliances load:

$$DL_{SA} = 9300 \text{ VA}$$

The total demand load for the neutral N:

$$DL_{tot,N} = 5,205 \text{ VA} + 2,355 \text{ VA} + 9,300 \text{ VA} = 16,860 \text{ VA}$$

Therefore, the demand load in terms of current for the neutral N:

$$I_{tot,N} = 16,860 \text{ VA}/240 \text{ V} = 70.3 \text{ A}$$

7.3.2 Optional Method for Sizing Service Entrance

As noted earlier, the size of phases for a dwelling service entrance size can be determined using the optional method. Indeed and when the ampacity of the service feeder conductors is larger than 100 A, the NEC allows the design to utilize the optional method for residential buildings served by 240/120 V or 208Y/120 V systems. It should be noted that the size of the neutral (N) needs to be still determined using the standard method. First, the general calculation procedure for the optional method is outlined. Then, an example is provided to clarify the application of the method to size the phases of a dwelling service entrance feeder.

The optional method to size the phase wires (i.e., A and B for one-phase systems and A, B, and C for three-phase systems) for the service entrance consists of the following steps:

1. *Demand load for general loads*:

   The demand load, $DL_{GL,O}$, for general loads for one or two-unit dwellings can be estimated as follows:

$$DL_{GL,O} = 100\% * \text{first } 10,000 \text{ VA} + 40\% * \left(CL_{GL,O} - 10,000 \text{ VA}\right) \tag{7.5}$$

   where $CL_{GL,O}$ are the connected loads associated with lighting, general-use receptacles, and appliances and are calculated as follows:

$$CL_{GL,O} = CL_{GL} + CL_{AP} \tag{7.6}$$

   where CL_{GL} is the connected general load associated with lighting and general-use receptacles and CL_{AP} is the connected load for all appliances except for space heating and cooling systems. The two connected loads are estimated as noted in the following sections.

   General loads (lighting/general receptacles)

   - General load = 3 VA/ft^2 (or 33 VA/m^2) * total livable building floor area
   - Kitchen load = 2 branch circuits * 1500 VA/circuit
     - Each branch circuit has to be rated at 20 A.
   - Laundry load = 1 branch circuit * 1500 VA/circuit
     - The branch circuit has to be rated at 20 A.

   By adding these three loads, the connected load associated with general lighting and general-use receptacles, CL_{GL}, can be determined.

   Appliances load:

   The connected load, CL_{AP}, for all appliances including fixed and special appliances, except space heating and cooling systems, considered in the standard method is determined by simply adding all the nameplate ratings for the appliances.

TABLE 7.3

Demand Factors for Heating and Air Conditioning Systems for Dwellings Specific to Optional Method

Type of Space Heating and Cooling Systems	Demand Factor (%)
Air conditioning and cooling system	100
Heat pump system without supplemental electric heating	100
Heat pump (HP) system with supplemental electric heating (H)	100 for HP 65 for H
Electrical heating system with less than four separately controlled units	65
Electrical heating system with four or more separately controlled units	40
Electrical thermal storage system	100

2. *Demand load for space heating and cooling systems*:

   The demand load, $DL_{HC,O}$, for the heating and air conditioning systems is the largest of the demand loads estimated using the demand factors and connect loads specific to various systems serving the dwelling, as noted in Table 7.3.

3. *Total demand load:* The total demand load, $DL_{tot\text{-}O}$, can be estimated by simply summing up the demand loads estimated in the previous three steps according to the following equation:

$$DL_{tot\text{-}O} = DL_{GL\text{-}O} + DL_{HC\text{-}O} \tag{7.7}$$

   Then, the total load can be expressed in terms of amperage depending on the line voltage for the service:

$$I_{tot} = \frac{DL_{tot\text{-}O}}{E_L} \text{ for one-phase systems} \left(\text{such as } 120/240 \text{ V}\right)$$
$$I_{tot} = \frac{DL_{tot\text{-}O}}{\sqrt{3}E_L} \text{ for three-phase systems} \left(\text{such as } 208\text{Y}/120 \text{ V}\right) \tag{7.8}$$

Example 7.3 illustrates the application of the optional method to size the service feeder for the same dwelling considered in Example 7.2.

Example 7.3

Problem

For the dwelling considered in Example 7.2, estimate the demand loads for phases A and B using the optional method instead of the standard method.

Solution

Under specific conditions, the NEC allows the use of the optional method to estimate the demand load for the phases A and B.
First all the rated loads (*except for the space heaters*) *are added*:

General load = 3 VA/ft^2 * 1600 ft^2 = 4800 VA
Kitchen load = 2 circuits * 1500 VA/circuit = 3000 VA
Laundry load = 1 circuit * 1500 VA/circuit = 1500 VA
Disposal = 400 VA

Compactor = 800 VA
Blower motor = 4.5 A * 120 V = 540 VA
Dishwasher = 1400 VA
Water heater = 4800 VA
Range = 12,000 VA
Dryer = 5000 VA

Thus, the total connect load for the dwelling without the space heaters:

$$CL_{tot\text{-}no\text{-}heaters} = 34,240\,\text{VA}$$

The total demand load for phases A and B using the optional method for all loads except the heaters:

$$DL_{tot,no\text{-}heaters} = 100\% * 10,000\,\text{VA} + 40\%\left(34,240 - 10,000\right)\text{VA} = 19,696\ \text{VA}$$

The additional load due to the space heaters (refer to Table 7.3 for the demand factor) is

$$DL_{heaters} = 20,000\,\text{VA} * 65\% = 13,000\,\text{VA}$$

Then, the total demand load for phases A and B using the optional method for all loads:

$$DL_{tot,no\text{-}heaters} = 19,696\,\text{VA} + 13,000\,\text{VA} = 32,696\,\text{VA}$$

Therefore, the demand load in terms of current for phases A and B using the optional method:

$$I_{tot,A\text{-}B} = 32,696\,\text{VA} / 240\,\text{V} = 136.2\,\text{A}$$

> In this example, the optional method results in lower demand loads than the standard method and thus can be used to justify reduced size for the service entrance for dwellings.

For dwellings with several units such as the case for apartment buildings, the demand factors provided in Table 7.4 should be considered to determine the size of the service entrance using the optional method as outlined in more details in Section 7.6.

7.3.3 Sizing Service Entrance

Based on the demand loads for the phases and the neutral, the size for the conductors as well as the main circuit breaker for the service entrance can be selected depending on the type of cable to be used (i.e., insulation layer and conductor temperature) as discussed in both Chapters 5 and 6. Table 7.5 can be used to determine the appropriate conductor sizes and circuit breaker ratings for dwellings with single-phase services. It should be noted that Section 310.15 of the NEC (2014) allows to adjust by a factor of 0.83 the reduce the size of 120/240 V service entrance ungrounded conductors for dwellings rated between 100 A through 400 A. The selection for both the conductor sizes and the main circuit breaker ratings are based on the demand load (in amps) for both the phases and neutral using either the standard or the optional method. It should be noted that for single-family residential buildings, the panelboard ratings (i.e., the main circuit breaker ratings) range from 100 to 200 A. For all residential buildings served by one-phase or three-phase systems, the sizes of conductors used should be based on the ampacity noted in Table 6.1 and the circuit breaker ratings listed in Table 5.4. Moreover and to enhance the safety of the electrical systems in residential buildings, the main panel has to be connected to the ground through a grounding conductor, as indicated in Figure 7.5. The size of the grounding conductor depends on

TABLE 7.4

Optional Method Demand Factors for Apartment Buildings

Number of Dwelling Units	Demand Factor
1–2*	100% for first 10 kVA 40% for remainder
2–3	45
6–7	44
8–10	43
11	42
12–13	41
14–15	40
16–17	39
18–20	38
21	37
22–23	36
24–25	35
26–27	34
28–30	33
31	32
32–33	31
34–36	30
37–38	29
39–42	28
43–45	27
46–50	26
51–55	25
56–61	24
Over 62	23

Note: This table is adapted from NEC-2014 Table 220.84 (NEC, 2014).

TABLE 7.5

Single-Phase Service Entrance Ratings and Feeder Conductor Sizes for Dwellings

	Service Entrance Conductor Size (AWG or MCM)	
Main Service Rating (A)	**Copper Wires**	**Aluminum or Copper-Clad Aluminum Wires**
100	2 AWG	1/0 AWG
125	1 AWG	2/0 AWG
150	1/0 AWG	3/0 AWG
175	2/0 AWG	4/0 AWG
200	3/0 AWG	250 MCM
225	4/0 AWG	300 MCM
250	250 MCM	350 MCM
300	350 MCM	500 MCM
350	500 MCM	700 MCM0
400	600 MCM or 2 parallel 3/0 AWG	Or 2 parallel 4/0 AWG 900 MCM Or 2 parallel 250 MCM

TABLE 7.6

Service Conductor Sizes for Dwellings

Service Entrance Conductor Size (AWG or MCM)		Service Grounding Conductor Size (AWG)	
Copper Wires	**Aluminum or Copper-Clad Aluminum Wires**	**Copper Wires**	**Aluminum or Copper-Clad Aluminum Wires**
3 or 2 AWG	1 or 1/0 AWG	8 AWG	6 AWG
1 or 1/0 AWG	2/0 or 3/0 AWG	6 AWG	4 AWG
2/0 or 3/0 AWG	4/0 AWG or 250 MCM	4 AWG	2 AWG
4/0 AWG – 350 MCM	300 – 500 MCM	2 AWG	1/0 AWG
400 – 600 MCM	600 – 800 MCM	1/0 AWG	3/0 AWG
700 – 1100 MCM	1000 – 1750 MCM	2/0 AWG	4/0 AWG

the size of the main service entrance feeder, as discussed in Chapter 6 and as summarized in Table 7.6. It should be noted that the size of the neutral should be larger than the grounding conductor. If the size of the grounding conductor is found to be larger than that of the neutral, the size of the neutral conductor should be increased to be the same as that of the grounding wire. More discussions on the grounding options are outlined in Section 250 of the NEC (2014).

Example 7.4 illustrates the selection approach for the service entrance conductors, main circuit breaker rating, and the grounding conductor.

Example 7.4

Problem

For the dwelling considered in Examples 7.2 and 7.3, size the service entrance conductor sizes as well as the main circuit breaker rating. Compare the service entrance sizes using the optional method instead of the standard method. Consider copper THW conductors for the service entrance.

Solution

Standard method:

Using the demand loads for the phases A and B as well as the neutral N found in Example 7.2 using the standard method, the rating of the panel is 200 A according to Table 7.5 (since the demand load is 184.8 A) and the size of the service entrance conductors for A and B is 3/0 THW (Cu) based on Table 7.5. Moreover, the neutral conductor size is 4 THW (Cu) based on Table 6.1 (since the demand load is 70.3 A). The grounding conductor size according to Table 7.6 is 4 AWG (Cu), which is the same as that of the neutral conductor.

Optional method:

Using the demand loads for the phases A and B found in Example 7.3 applying the optional method, the rating of the panel can be reduced to 150 A according to Table 7.5 (since the demand load is 136.2 A) and the size of conductors for A and B is 1 (based on Table 7.4). The grounding conductor size according to Table 7.6 is 6 AWG (Cu), which is lower than that of the neutral conductor found to be 4 THW (Cu) using the standard method calculations.